

SUKUMAR BV

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SUMMARY

AI & Machine Learning Engineering student (CGPA: 8.97/10) with experience building and deploying end-to-end machine learning applications. Skilled in Python, FastAPI, PyTorch, TensorFlow, and Docker, with project experience spanning NLP, Computer Vision, MLOps, and predictive analytics. Passionate about turning data-driven ideas into practical solutions and eager to contribute to real-world AI products while continuing to grow as an engineer.

SKILLS

Languages: Python, Java, JavaScript, SQL, C

AI / Machine Learning: NLP, Text-to-SQL, Computer Vision, LSTM, Scikit-learn, PyTorch, TensorFlow, CVZone

Frameworks & Libraries: FastAPI, React, SQLAlchemy, OpenCV, SciPy, PyAutoGUI

Databases & DevOps: PostgreSQL, Docker, Docker Compose

Developer Tools: Git & GitHub, VS Code, ESP32-S3

Soft Skills: Communication, Problem Solving, Team Collaboration, Research and Documentation

PROJECTS

AI-Powered Air Quality Monitoring System (ISRO Project)

Nov 2025 – Present

Python, LSTM, SciPy, C++, ESP32-S3

- **Role & Scope:** Collaborating as a System Engineer and Data Analyst Project Intern to develop a next-generation edge-computing AQMS for the Indian Space Research Organisation (ISRO).
- **ML Forecasting:** Co-developing a hybrid temporal forecasting pipeline with an LSTM network that ingests physics-validated outputs to predict real-time air quality and automate regulatory compliance tracking.
- **Physics-Based Modeling:** Assisting in the migration from a legacy PINN to a deterministic Chemical Transport Model (CTM), applying operator splitting for cleaner atmospheric data simulation.
- **Edge Systems:** Integrating an ESP32-S3 microcontroller with I2C/UART expansion bridges to ensure zero-loss, non-blocking sensor data feeds for the ML pipeline.

ModelOps Studio | [Live Demo](#)

June 2026

Python, FastAPI, Scikit-learn, SQLAlchemy, Docker, Plotly, Jinja2, SQLite

- Architected a lightweight, Vertex AI-inspired ModelOps platform covering the complete ML lifecycle, from dataset ingestion and exploration to training, comparison, deployment, and monitoring, exposed through a server-rendered UI and a documented FastAPI JSON API.
- Implemented automated training and evaluation across five algorithms with a shared preprocessing pipeline, 3-fold cross-validation, and dynamic classification/regression detection, versioning every run with metrics, hyperparameters, and training logs.
- Built a monitoring service using Kolmogorov-Smirnov and Population Stability Index tests to detect data drift against training distributions and surface retrain recommendations.
- Containerized the full application with Docker and deployed it on Render, backed by a SQLAlchemy ORM layer portable across SQLite and PostgreSQL.

NLP Query Engine | [Live Demo](#)

Jan 2026

Python, FastAPI, React, PostgreSQL, Docker

- Architected a full-stack natural-language-to-SQL engine with three entry points, PostgreSQL connection, file upload, and an offline demo dataset, unified behind a pluggable data-source abstraction.
- Implemented dynamic schema discovery that introspects any connected SQL backend, including CSV and XLSX worksheets loaded into an in-memory SQLite database, with no hard-coded schema definitions.
- Built a hybrid query classifier that routes each question to SQL generation, retrieval-augmented document search, or both, combining LLM-based Text-to-SQL with embedding-based retrieval over ingested PDFs, DOCX, and TXT files.
- Engineered a single-container production build that serves the compiled React frontend and FastAPI backend from one port, containerized with Docker and deployed on Render.

Product Recommendation System

Dec 2025

Python, Scikit-learn, Pandas

- Designed dual recommendation pipelines combining content-based and collaborative filtering to compare item-similarity and user-behavior-driven approaches on the same catalog.
- Engineered TF-IDF feature extraction with cosine similarity indexing over item descriptions and categories to generate product-to-product recommendations.

CERTIFICATIONS

Artificial Intelligence: Knowledge Representation and Reasoning

Gen AI Powered Data Analytics Job Simulation

Computer Graphics

NPTEL

TATA iQ, Forage

NPTEL

EDUCATION

Cambridge Institute of Technology

Bachelor of Engineering (AI & ML)

2023 – 2027

CGPA: 8.97/10